Check for updates

Received: 6 May 2023 | Revised: 22 June 2023 | Accepted: 24 July 2023

DOI: 10.1049/rpg2.12814

IET Renewable Power Generation

The Institution of Engineering and Technology WILEY

ORIGINAL RESEARCH

Active and reactive power coordinated optimization of active distribution networks considering dynamic reconfiguration and SOP

Chen Li | **Yao Dai** | **Peng Wang** | **Shiwei Xia**

School of Electrical and Electronic Engineering, North China Electric Power University, Beijing, China

Correspondence
Shiwei Xia, School of Electrical and Electronic Engineering, North China Electric Power University, Beijing, China.
Email: s.w.xia@connect.polyu.hk

Funding information
State Grid Corporation of China, the Science & Technology project: Research and application of hierarchical and collaborative regulation and control technology of medium and low voltage distribution network for photovoltaic access in the whole county, Grant/Award Number: 5400-202216414A-2-0-ZN

Abstract

In this paper, a day-ahead active and reactive power coordinated optimization strategy for active distribution networks (ADNs) with dynamic reconfiguration and soft open point (SOP) is proposed. Firstly, the model of dynamic reconfiguration of distribution networks and SOP are established. Afterward, the day-ahead active and reactive power coordinated optimization model of ADNs is established to minimize ADNs daily operation cost with consideration of the operation constraints of distributed generation, reactive power compensation equipment, dynamic reconfiguration of distribution networks and SOP. Then the model is converted into a mixed-integer second-order cone programming problem by using the Big-M method and the second-order cone relaxation and afterward is solved by YALMIP and CPLEX. Finally, the effectiveness of the proposed second-order cone optimization model is validated by a modified IEEE 33-node power distribution network.

1 | INTRODUCTION

In recent years, traditional distribution networks have been gradually transformed into active distribution networks (ADNs) due to the high level of distributed power sources (DGs), such as the large-scale photovoltaic (PV), wind turbines (WT), micro gas turbine (MGT), and energy storage systems (ESSs) [1, 2]. Traditional dispatching methods are difficult to adapt to the complex and frequent regulation needs of ADN [3]. Nowadays, the focus of distribution network research is to exploit the diverse regulation potential of ADNs [3] and to improve their operation economics and safety through coordinated optimization of multiple devices in ADNs [4, 5].

Dynamic reconfiguration of distribution networks is a typical active management technology to effectively reduce network losses and improve system voltage levels. The dynamic reconfiguration of distribution networks has been widely investigated in previous studies on the day-ahead operation with the objectives of improving renewable energy consumption and reducing network loss by mathematical optimization methods [6, 7] or heuristic algorithms [8, 9]. Furthermore, a few studies considered the impact of smart soft open point (SOP) in distribution network reconfiguration. Authors of ref. [10] performed network reconfiguration by adjusting SOP devices and effectively reducing the system network loss. However, the voltage problem was not considered in the optimization. Ref. [11] further considered the DG output limits and its operation constraints, but the reactive power compensation capability of DG was ignored. Ref. [12] pointed out that the coordination of SOP and dynamic reconfiguration would properly regulate voltage profiles of distribution networks when optimizing the network loss. These studies only focused on active power optimization and did not consider reactive power/voltage optimization. Only distributed power supplies were modelled while multiple reactive power/voltage control devices were ignored in ADNs.

Only a few studies have investigated the coordinated optimization of reactive power/voltage control devices, distributed power supplies and network dynamic reconfiguration technologies at the day-ahead scale. In one such study [13], a network reconfiguration method considering reactive/voltage control devices, such as static var compensator (SVC) and discrete capacitor banks (CB) with distributed renewable

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial-NoDerivs License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.
© 2023 The Authors. *IET Renewable Power Generation* published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology.

energy sources, was proposed. The problem was converted to a mixed integer convex programming model and then solved. Ref. [14] further enriched the variety of reactive/voltage control devices for active and reactive power coordinated optimization to minimize the operating cost. A certain degree of optimization to the node voltages was achieved in the two aforementioned studies, but neither of them considered ESS and MGT. In Ref. [15], the authors achieved performance enhancement of the distribution system based on an improved tunicate swarm algorithm that carried out coordinated optimization of network reconfiguration with multiple controllable resources. Nonetheless, the reactive/voltage control devices consisted only of CBs, and problems were associated with the local convergence of the heuristic algorithm. All of the above studies failed to completely model DG and reactive/voltage control devices and did not consider the role of SOP devices in the optimization of distribution system operation.

Under this context, a day-ahead active and reactive power coordinated optimization for ADNs considering network reconfiguration and SOP is proposed in this paper. In this research, the distribution system operation economy and voltage quality have been improved through the coordinated optimization of network dynamic reconfiguration, SOP operation, distributed power supply output scheduling, and reactive-voltage control device regulation. In terms of the solution algorithm, the optimal scheduling model has been transformed into a mixed-integer second-order cone programming (MISOCP) problem [16] and solved on the YALMIP platform using the CPLEX solver.

The contributions of this paper are twofold.

1. A comprehensive set of power regulation devices including MGTs and ESSs, SVCs, CBs, switches, and SOPs are fully exploited for the active-reactive collaboratively optimization of active distribution networks, compared to the separately active or reactive power optimizing in refs. [11] and [13], thereby the proposed model effectively reduces ADN operating cost and improve system nodal voltage profiles simultaneously.
2. Effectiveness of the proposed optimization model is validated by the simulation of a modified active distribution network, and the impacts of ESS, MGT, dynamic network reconfiguration by switches and SOPs on ADN operating cost and nodal voltage promotion are fully analyzed and discussed.

The rest of this paper is organized as follows. The models of dynamic reconfiguration and SOP are presented in Section 2, and the day-ahead active and reactive power coordinated optimization of the distribution network is proposed in Section 3. Afterward, a case study and discussion are conducted in Section 4. The conclusion is drawn in the last section.

2 | THE OPERATION MODEL OF DYNAMIC NETWORK RECONFIGURATION AND SOP

2.1 | Dynamic reconfiguration model of distribution networks

Dynamic reconfiguration of a distribution network refers to keeping the distribution network in the most appropriate network topology at different time periods by opening/closing switches or controlling SOP devices, so as to reduce power network loss, optimize system voltage, and improve the operating economy of a distribution network [17]. The time resolution of day-ahead optimization is typically 1 h, while dynamic reconfiguration is also an hourly-adjusted regulation strategy for the distribution network operation and thus is compatible with day-ahead optimization. In this paper, the following two typical constraints are applied when performing dynamic reconfiguration: distribution network radial topology and switch operation constraints.

1. Distribution network radial topology constraint

The distribution network topology is radial and it is necessary to ensure that there are no loops and islands in the distribution network when dynamic reconfiguration is performed [18]. The constraints can be expressed using the following Equations (1)-(3):

$$\sum_{ij\in\psi_b} \alpha_{ij,t} = N - 1 \tag{1}$$

$$\beta_{ij,t} + \beta_{ji,t} = \alpha_{ij,t} \tag{2}$$

$$\sum_{i=1}^{N} \beta_{ij,t} = 1 \qquad \forall j \in \psi_{n-1} \tag{3}$$

where $\alpha_{ij,t}$ is a binary variable indicating whether the branch ij is open or closed at time t. If the value is 1, the branch ij is closed at time t, and the branch ij isn't closed if the value is 0; N is the number of nodes in the distribution network; $\beta_{ij,t}$ is a 0–1 variable and $\beta_{ij,t} = 1$ indicates that node i is the parent of node j at time t; and Ψ_{n-1} represents the set of nodes in the distribution network excluding the root node.

2. Operation constraint of switches

The switches in the distribution network could flexibly adjust the distribution network topology via opening or closing the contact points, but frequent switching operations would seriously reduce the lifespan of the switches. Moreover, frequent topology changes of the distribution network could deteriorate the operation reliability of a distribution system. Therefore, it is necessary to restrict the maximum operation number of one switch, the minimum operation interval of one switch, and the

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

total operation number of all switches per day in the day-ahead optimization.

$$\sum_{t=1}^{T-1} |\alpha_{ij,t+1} - \alpha_{ij,t}| \le n_i^{\text{total}} \qquad \forall(\text{i, j}) \,|ij \in \psi_b \tag{4}$$

$$\sum_{a=1}^{m} |\alpha_{ij,t+a} - \alpha_{ij,t+a-1}| \le 1 \tag{5}$$

$$\sum_{t=1}^{T-1} \sum_{ij \in \psi_b} |\alpha_{ij,t+1} - \alpha_{ij,t}| \le n_s^{\text{total}} \tag{6}$$

where n_i^{total} is the maximum operation number of a single switch per day; n_s^{total} is the maximum operation number per day for all switches per day; and m is the minimum number of hours for the state change of one switch.

2.2 | SOP operation model

SOP is a new type of power electronic device installed in distribution networks and is usually used as a replacement for traditional contact switches to achieve flexible "soft connection" of feeders and provide accurate, fast, and flexible power exchange control [19]. The installation location of SOPs is shown in Figure A1 in Appendix A. Precise control of active and reactive power can be achieved by SOPs with real-time dispatching strategies, thereby enabling excellent power flow control and voltage control capability.

SOP devices could be used for distribution network operation optimization in short time scales such as in seconds/minutes as well as in hourly long time scales. From the perspective of day-ahead optimization, SOPs can be regulated at the hourly scale to modify the connection state of branches, thereby engaging in distribution network dynamic reconfiguration. In contrast, to contact switches based on the open/close binary states, SOPs can be used to conduct the continuous and precise power flow control by open/close binary states and the continuous power regulating capability. Coupling SOP with contact switches for distribution network reconfiguration could effectively reduce system network loss, improve distribution system operating economy and promote voltage level [20, 21]. SOP devices are subject to the following active power constraints and capacity constraints at both ends in normal conditions [22].

1. Active power flow constraint of SOP

$$P_i^{\text{SOP}} + P_j^{\text{SOP}} + P_i^{\text{SOP},L} + P_j^{\text{SOP},L} = 0 \tag{7}$$

$$P_i^{\text{SOP},L} = \eta^{\text{SOP}} \sqrt{\left(P_i^{\text{SOP}}\right)^2 + \left(Q_i^{\text{SOP}}\right)^2} \tag{8}$$

$$P_j^{\text{SOP},L} = \eta^{\text{SOP}} \sqrt{\left(P_j^{\text{SOP}}\right)^2 + \left(Q_j^{\text{SOP}}\right)^2} \tag{9}$$

where P_i^{SOP} and Q_i^{SOP} are the active and reactive power at node i of SOP, respectively; $P_j^{\text{SOP},L}$ is the active power loss at node i of a converter; and η^{SOP} is the loss factor of the converter.

2. Thermal capacity constraint

$$\sqrt{\left(P_i^{\text{SOP}}\right)^2 + \left(Q_i^{\text{SOP}}\right)^2} \le S_i^{\text{SOP}} \tag{10}$$

$$\sqrt{\left(P_j^{\text{SOP}}\right)^2 + \left(Q_j^{\text{SOP}}\right)^2} \le S_j^{\text{SOP}} \tag{11}$$

where S_i^{SOP} and S_j^{SOP} are the thermal capacities at nodes i and j of a converter.

3 | PROPOSED DAY-AHEAD COORDINATED OPTIMIZATION MODEL OF ADN

3.1 | Coordinated optimization objective of ADN

In this paper, the objective is to fully exploit the power regulation potential of ADN and enhance its operation economy and security by using SOP devices and dynamic reconfiguration technology. To achieve this, an ADN day-ahead active and reactive power coordinated optimal operation model is proposed with the hourly resolution for a total of 24 h. The objective is to minimize the overall daily operating cost of the distribution system, including the power purchase cost, MGT operation cost, and voltage deviation penalty cost as shown in Equation (12).

$$\min C = \text{C}_{\text{sup}} + C_{\text{MGT}} + C_{\text{DV}} \tag{12}$$

$$\begin{cases} \text{C}_{\text{sup}} = \sum_{t=1}^{T} c_{\text{sup},t} P_{\text{sup},t} \cdot \Delta T \\ C_{\text{MGT}} = \sum_{i \in \psi_{\text{MGT}}} \sum_{t=1}^{T} \left(a_i P_{i,t}^{\text{MGT}} + b_i \mu_{i,t}^{\text{MGT}} \right. \\ \qquad \left. + c_{i,s}^{\text{MGT}} \mu_{i,t}^{\text{MGT}} \left(1 - \mu_{i,t-1}^{\text{MGT}}\right) + c_{i,q}^{\text{MGT}} \mu_{i,t-1}^{\text{MGT}} \left(1 - \mu_{i,t}^{\text{MGT}}\right) \right) \\ C_{\text{DV}} = c_{\text{dv}} \sum_{i=1}^{N} \sum_{t=1}^{T} \left| U_{i,t}^2 - U_{i,e}^2 \right| \end{cases} \tag{13}$$

where C is the total operating cost; C_{sup} is the electricity cost purchased from the upstream power grid; C_{MGT} is the MGT operating cost; C_{DV} is the voltage deviation penalty cost; T is the total number of dispatching periods; ΔT is the length of a single period; ψ_{MGT} is the set of all MGTs in the network; $c_{\text{sup},t}$ and $P_{\text{sup},t}$ are the price and power purchased from the upstream power grid at time t, respectively; $\mu_{i,t}^{\text{MGT}}$ is the 0–1 variable representing the on-off state of the ith MGT at time t, and $\mu_{i,t}^{\text{MGT}} = 1$ when MGT i is online at time t; $P_{i,t}^{\text{MGT}}$ is the active output value of the ith MGT at time t; a_i and b_i are the cost coefficients of the ith MGT, respectively; $c_{i,s}^{\text{MGT}}$ and $c_{i,q}^{\text{MGT}}$

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

are the start and stop cost of the ith MGT, respectively; $U_{i,t}$ is the voltage magnitude of node i at time t; $U_{i,e}$ is the desired voltage of node i, which usually assumes as the rated voltage; and c_{dv} is the voltage deviation penalty factor considered as 5 Yuan/(p. u.)^2 in this paper.

3.2 | Operating constraints of ADN coordinated optimization model

1. Distflow constraint of ADNs

The distflow model is derived from branch power equations and is widely used in the power flow calculation of radial distribution networks [23]. In this paper, an improved distflow model is established with consideration of the network topology dynamic reconfiguration. This model is applicable for varied branch states by introducing the line state variable $\alpha_{ij,t}$ and relaxing the current constraint via the big M method.

$$U_{i,t}^{s} = U_{i,t}^{2} \tag{14}$$

$$I_{ij,t}^{s} = I_{ij,t}^{2} \tag{15}$$

$$\sum_{i\in f(j)} P_{ij,t} - r_{ij} I_{ij,t}^{s} = \sum_{k\in s(j)} P_{jk,t} + P_{j,t} \tag{16}$$

$$\sum_{i\in f(j)} Q_{ij,t} - x_{ij} I_{ij,t}^{s} = \sum_{k\in s(j)} Q_{jk,t} + Q_{j,t} \tag{17}$$

$$-\alpha_{ij,t} M_1 \le P_{ij,t} \le \alpha_{ij,t} M_1 \tag{18}$$

$$-\alpha_{ij,t} M_2 \le Q_{ij,t} \le \alpha_{ij,t} M_2 \tag{19}$$

$$-\alpha_{ij,t} M_3 \le I_{ij,t} \le \alpha_{ij,t} M_3 \tag{20}$$

$$\begin{aligned} &U_{j,t}^{s} \le M_4(1-\alpha_{ij,t}) + U_{i,t}^{s} \\ &-2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + \left(r_{ij}^2 + x_{ij}^2\right) I_{ij,t}^{s} \end{aligned} \tag{21}$$

$$\begin{aligned} &U_{j,t}^{s} \ge -M_4(1-\alpha_{ij,t}) + U_{i,t}^{s} \\ &-2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) + \left(r_{ij}^2 + x_{ij}^2\right) I_{ij,t}^{s} \end{aligned} \tag{22}$$

$$U_{i,t}^{s} I_{ij,t}^{s} = P_{ij,t}^{2} + Q_{ij,t}^{2} \forall (i,j) \,|\, ij \in \psi_b \tag{23}$$

where $I_{ij,t}$, $P_{ij,t}$, and $Q_{ij,t}$ are the current, active power, and reactive power through the branch ij at time t, respectively; r_{ij} and x_{ij} are the resistance and reactance of the branch ij; $f(j)$ and $s(j)$ are the set of parent and child nodes of node j, respectively; M_1 to M_4 are positive and sufficiently large numbers. $P_{j,t}$ and $Q_{j,t}$ are the active power and reactive power injected into node j at time t, respectively, which could be calculated as follows.

$$P_{j,t} = P_{j,t}^{f} + P_{j,t}^{wind} + P_{j,t}^{pv} + P_{j,t}^{MGT} + P_{j,t}^{dis} - P_{j,t}^{ch} - P_{j,t}^{load} \tag{24}$$

$$Q_{j,t} = Q_{j,t}^{f} + Q_{j,t}^{svc} + Q_{j,t}^{cb} + Q_{j,t}^{MGT} - Q_{j,t}^{load} \tag{25}$$

where $P_{j,t}^{wind}$ and $P_{j,t}^{pv}$ are the wind power and photovoltaic power generation injected into node j at time t, respectively; $P_{j,t}^{load}$ and $Q_{j,t}^{load}$ are the active and reactive power of node j at time t; $P_{j,t}^{dis}$ and $P_{j,t}^{ch}$ are the discharging and charging power of an ESS connected at node j at time t, respectively; and $P_{j,t}^{f}$ and $Q_{j,t}^{f}$ are the active and reactive power of node j purchased from the upstream power grid at time t.

2. Nodal voltage and branch thermal capacity constraints

To ensure the operation security of the distribution network, the nodal voltage and branch thermal capacity are constrained as follows:

$$\left(U_i^{min}\right)^2 \le U_{i,t}^{s} \le \left(U_i^{max}\right)^2 \tag{26}$$

$$0 \le I_{ij,t}^{s} \le \left(I_{ij}^{max}\right)^2 \tag{27}$$

where U_i^{min} and U_i^{max} are the minimum and maximum voltages of node i; I_{ij}^{max} is the maximum current allowed for branch ij.

3. Operation constraints of MGT

MGT is used to realize the decoupled optimization of active and reactive power of the distribution network, by the upper and lower limits and apparent power constraints [24] in Equations (28) and (29) respectively.

$$\begin{cases} \mu_{i,t}^{MGT} P_{MGTi}^{min} \le P_{i,t}^{MGT} \le \mu_{i,t}^{MGT} P_{MGTi}^{max} \\ \mu_{i,t}^{MGT} Q_{MGTi}^{min} \le Q_{i,t}^{MGT} \le \mu_{i,t}^{MGT} Q_{MGTi}^{max} \end{cases} \tag{28}$$

$$\begin{cases} \sqrt{\left(P_{i,t}^{MGT}\right)^2 + \left(Q_{i,t}^{MGT}\right)^2} = S_{i,t}^{MGT} \\ S_{i,t}^{MGT} \le S_{MGTi}^{max} \end{cases} \tag{29}$$

where $Q_{i,t}^{MGT}$ and $S_{i,t}^{MGT}$ are the reactive power and apparent power of ith MGT at time t; P_{MGTi}^{min} and P_{MGTi}^{max} are the minimum and maximum active power output of the ith MGT; Q_{MGTi}^{min} and Q_{MGTi}^{max} are the minimum and maximum reactive power output of the ith MGT, respectively; S_{MGTi}^{max} is the maximum apparent power of the ith MGT.

4. Renewable power generation constraint

The distributed wind power and photovoltaic power (PV) generation in a distribution system should meet the constraints of Equations (31) and (32).

$$P_{WTi}^{min} \le P_{i,t}^{wind} \le P_{WTi}^{max} \tag{30}$$

$$P_{PVi}^{min} \le P_{i,t}^{pv} \le P_{PVi}^{max} \tag{31}$$

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

where P_{WTi}^{min} and P_{WTi}^{max} are the minimum and maximum wind power output of node i at time t; P_{PVi}^{min} and P_{PVi}^{max} are the minimum and maximum photovoltaic power output of node i at time t, respectively.

5. Operation constraint of ESS

The operation constraints of ESS include the mutual exclusion condition of the charging and discharging state, the energy capacity constraints, and charging/discharging power output limits [25], as shown in Equations (32)–(34).

$$\alpha_{i,t}^{ch} + \alpha_{i,t}^{dis} \le 1 \tag{32}$$

$$\begin{cases} 0 \le P_{i,t}^{ch} \le \alpha_{i,t}^{ch} P_{ESSi}^{max} \\ 0 \le P_{i,t}^{dis} \le \alpha_{i,t}^{dis} P_{ESSi}^{max} \end{cases} \tag{33}$$

$$\begin{cases} E_{i,t+1}^{ESS} = E_{i,t}^{ESS}(1-\eta) + \eta^{ch} P_{i,t}^{ch} \Delta T - \frac{1}{\eta^{dis}} P_{i,t}^{dis} \Delta T \\ 0.2E_{ESSi}^{max} \le E_{i,t}^{ESS} \le 0.9E_{ESSi}^{max} \end{cases} \tag{34}$$

where $\alpha_{i,t}^{ch}$ and $\alpha_{i,t}^{dis}$ are the charging and discharging state variables of energy storage i at time t, and $\alpha_{i,t}^{ch} = 1$ indicates the ith ESS in charging state, and $\alpha_{i,t}^{dis} = 1$ indicates the ith ESS in discharging state; P^{max} is the maximum charging or discharging power of the ith ESS; E^{max} is the maximum capacity of the ith ESS; η^{ch} and η^{dis} are the charging and discharging efficiency of ESS; η is the self-discharge rate of the ESS.

6. Operation constraint of SVC

The reactive power output limits of SVC are modelled by Equation (35).

$$Q_{svci}^{min} \le Q_{i,t}^{svc} \le Q_{svci}^{max} \tag{35}$$

where Q_{svci}^{min} and Q_{svci}^{max} are the maximum and minimum reactive output of the ith SVC device, respectively.

7. Operation constraint of CB

The CB operation constraints include the maximal reactive power output of CB, and the maximal number of changed CB capacity [12] as shown in Equations (36)–(38).

$$Q_{i,t}^{cb} = y_{i,t}^{cb} Q_{cbi}^{step} \tag{36}$$

$$0 \le y_{i,t}^{cb} \le Y_{cbi}^{max} \tag{37}$$

$$\sum_{t=1}^{T} \left| y_{i,t}^{cb} - y_{i,t-1}^{cb} \right| \le N_{cbi}^{max} \tag{38}$$

where $Q_{i,t}^{cb}$ is the reactive power of the ith CB at time t; $y_{i,t}^{cb}$ is the number of the ith CB at time t; Q_{cbi}^{step} is the basic compensation power of the ith CB; Y_{cbi}^{max} is the maximum number of the ith CB; and N_{cbi}^{max} is the maximal changed number of the ith CB.

8. Operation constraint of OLTC

The online tap changer (OLTC) operation constraints include the upper and lower voltage regulation limits and the maximum number of OLTC tap ratio changed during a day [26], as shown in Equations (39)–(41):

$$U_n^{min} \le U_n^e R_t \le U_n^{max} \tag{39}$$

$$\underline{R} \le R_t \le \overline{R} \tag{40}$$

$$\sum_{t=1}^{T-1} \mu_t \le N_o^{total} \tag{41}$$

where R_t is a discrete operating ratio of the OLTC, i.e. the voltage of the secondary side over that of the primary side; $\bar{R}$ and are the upper and lower limits of tap ratio; μ_t is a binary state variable indicating whether the OLTC tap ratio is changed for continuous time periods, i.e. if $R_{t+1} - R_t = 0$ μ_t is 0, otherwise μ_t is 1; N_o^{total} is the maximum number of OLTC tap ratio changed during a day.

9. Operation constraints of dynamic reconfiguration

The constraints of distribution network dynamic reconfiguration are shown in Equations (1)–(6).

10. Operation constraint of SOPs

The operation constraints of SOPs are shown in Equations (7)–(11).

3.3 | Second-order cone transformation of the proposed model

The above-mentioned distribution network distflow constraints, SOP and MGT operation constraints are non-linear non-convex and thus are difficult to be solved directly by off-the-shelf solvers such as CPLEX. A second-order cone strategy is used in this paper to transform the non-convex constraints into a MISOCP problem, which could be readily solved by a commercial solver. The second-order cone optimization method calculates the optimization direction and step size based on linear transformation and dual theory thus can achieve fast convergence and ensure the solution optimality of the original optimization problem.

1. Conversion of distflow constraints

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Parameters of MGTs.

MGT	Output limits (MW)		Cost coefficients (Yuan)			
	P_{MGTi}^{min}	P_{MGTi}^{max}	a_i	b_i	$c_{i,s}^{MGT}$	$c_{i,q}^{MGT}$
At Node 3	0.1	0.5	510	70	400	200
At Node 9	0.2	0.6	440	145	750	400

The aforementioned distribution network distflow constraint (23) is non-convex and could be substituted with Equation (42) by using performing the second-order cone relaxation [27].

$$P_{ij,t}^2 + Q_{ij,t}^2 \le U_{i,t}^s I_{ij,t}^s \qquad \forall (i,j) \,|\, ij \in \psi_b \tag{42}$$

2. Conversion of SOP operation constraints

The active power flow constraints (8) and (9) of SOP are converted into the second-order cone inequalities by Equations (43) and (44).

$$\left(P_{i,t}^{SOP}\right)^2 + \left(Q_{i,t}^{SOP}\right)^2 \le \left(\frac{P_{i,t}^{SOP,L}}{\eta^{SOP}}\right)^2 \tag{43}$$

$$\left(P_{j,t}^{SOP}\right)^2 + \left(Q_{j,t}^{SOP}\right)^2 \le \left(\frac{P_{j,t}^{SOP,L}}{\eta^{SOP}}\right)^2 \tag{44}$$

3. Conversion of MGT operation constraints

The apparent power constraint (29) of MGT operation is converted into a second-order cone inequality as Equation (45).

$$\left(P_{i,t}^{MGT}\right)^2 + \left(Q_{i,t}^{MGT}\right)^2 \le \left(S_{MGTi}^{max}\right)^2 \tag{45}$$

4. Conversion of voltage deviation penalty cost

Two sets of non-negative auxiliary variables are introduced to convert the absolute value of the voltage deviation penalty cost (13) into Equation (46) with a set of linear constraints (47).

$$C_{DV} = c_{dv} \sum_{i \in N} \sum_{t=1}^{T} \left(S_{i,t}^a + S_{i,t}^b\right) \tag{46}$$

$$\begin{cases} -S_{i,t}^a \le U_{i,t}^2 - U_{i,e}^2 \le S_{i,t}^b \\ S_{i,t}^a \ge 0 \\ S_{i,t}^b \ge 0 \end{cases} \tag{47}$$

Based on these transformations, the proposed day-ahead active and reactive power coordinated optimization model of ADN with consideration of the network dynamic reconfiguration and SOP, which is non-linear and non-convex, can be fully expressed as a convex optimization problem as below.

$$\text{Min } C$$

$$\text{s.t.} \begin{cases} \text{Power flow constraint :} \\ \text{Equations (14)–(22), (24), (25), and (42)} \\ \text{Node voltage and circuit constraint :} \\ \text{Equations (26) and (27)} \\ \text{MGT operation constraint :} \\ \text{Equations (28), (29), and (45)} \\ \text{New energy output constraint :} \\ \text{Equations (30) and (31)} \\ \text{ESS constraint : Equations (32)–(34)} \\ \text{SVC constraint : Equation(35)} \\ \text{CB constraint : Equations (36)–(38)} \\ \text{Main transformer OLTC constraint :} \\ \text{Equations (39)–(41)} \\ \text{Distribution network reconfiguration constraint :} \\ \text{Equations (1)–(6)} \\ \text{SOP constraint :} \\ \text{Equations (7), (10), (11), (43), and (44)} \\ \text{Objective function conversion constraint :} \\ \text{Equations (46) and (47)} \end{cases}$$

4 | CASE STUDY AND RESULTS DISCUSSION

4.1 | Parameters of case study

The modified IEEE 33-node distribution power system shown in Figure 1 was used to verify the effectiveness of the proposed model. The system had 33 nodes with 37 branches, a reference voltage of 12.66 kV, and a base power of 1 MVA. The allowed maximum and minimum voltages were 1.05 and 0.95 p.u., respectively, and the maximum current allowed in the branches was 800 A. A constant power factor of 0.95 was used for both wind power and PV generation, and their predicted value is shown in Figure A2 in Appendix A, which approximately takes a penetration level of 60% of the total load demand. Two MGTs were connected at nodes 3 and 9, and the MGT parameters are shown in Table 1 [28]. An ESS was connected at node 8 with a capacity of 5 MWh, and its maximum charge/discharge power is 400 kW, with a self-discharge rate of 0.99 [29]. Five candidate groups of CBs were connected at node 26 with one group capacity of 100 kVA. One SVC was connected at node 17 with a capacity of 500 kVA. The SOP was connected to branches 8–14 with a capacity of 600 kVA and a loss factor of 0.01 [20]. The maximum number of OLTC tap ratio changed during a day was four. A time-of-use price as shown in Table 2, which is settled based on the tariff of general commercial and industrial customers in Beijing, is used for the electricity purchased cost in (13). Regarding

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

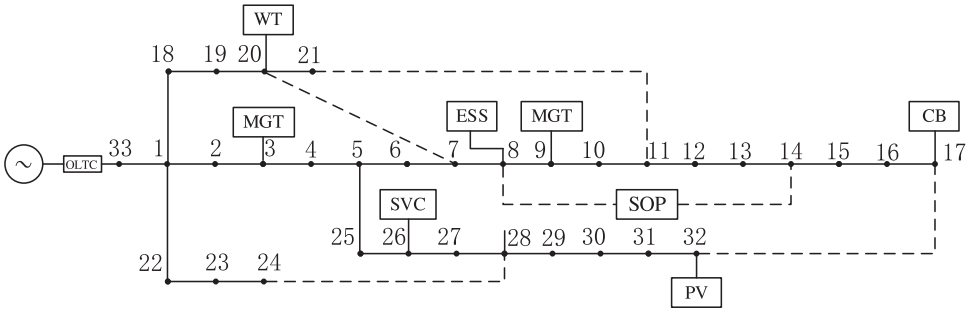


FIGURE 1 Modified IEEE 33-bus distribution system

TABLE 2 Time-of-use electricity price.

Time periods	Price (Yuan/MWh)
0:00–6:00, 22:00–24:00	400
6:00–8:00, 12:00–18:00	800
8:00–12:00, 18:00–22:00	1200

TABLE 3 Network dynamic reconfiguration.

Time periods	Disconnected switches
0:00–3:00	2–3, 10–11, 11–12, 17–18, 30–31
3:00–6:00	2–3, 10–11, 11–12, 27–28, 30–31
6:00–9:00	2–3, 10–11, 11–12, 14–15, 27–28
9:00–12:00	2–3, 8–9, 11–12, 14–15, 18–19
12:00–18:00	2–3, 5–6, 8–9, 11–12, 18–19
18:00–21:00	2–3, 8–9, 11–12, 18–19, 31–32
21:00–24:00	8–9, 11–12, 18–19, 21–22, 31–32

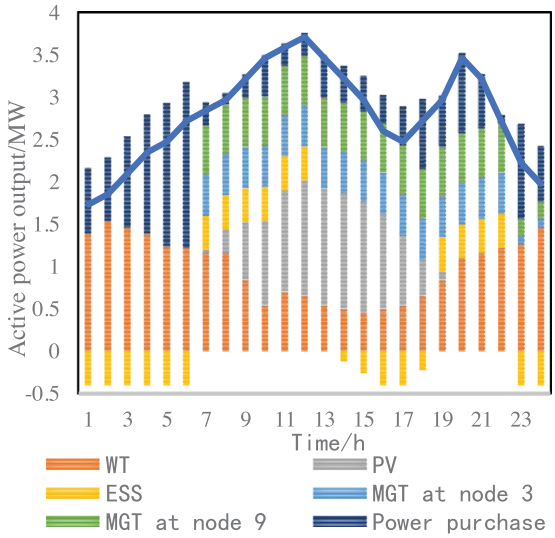


FIGURE 2 Active power in day-ahead scheduling

network reconfiguration, one reconfiguration period was set as 3 h. The maximum operation number of one switch, the minimum operation interval of one switch, and the total operation number of all switches are respectively fixed as 3, 2, and 15 per day. In this paper, the CPLEX solver was combined with the Yalmip toolkit in the Matlab 2016b environment to solve the problem.

4.2 | General optimization results

The distribution network dynamic reconfiguration results are shown in Table 3, where the total number of on-off states changed for contact switches was 14, and consequently 7 different network topologies are obtained throughout the day. The daily operating cost of the distribution network was 24,249 Yuan, of which the cost of purchasing power from the upstream power grid was 10,811 Yuan, the MGT power generation cost was 13,389 Yuan, and the system voltage penalty cost was 48 Yuan. The amount of power consumed due to network loss in the distribution network was 0.877 MWh.

The day-ahead optimal active power outputs are shown in Figure 2. The ESS was on charge state during 23:00–6:00 and 14:00–18:00, and discharged during 7:00–13:00 and 19:00-22:00. Both MGTs were started at 06:00 and operated at the minimum output after 22:00. The wind power and PV generation are operated at the maximum output throughout the day. Based on the optimized generation of ESS and MGTs, optimally purchased power from the upstream power grid, and wind power and PV generation, the total load is effectively balanced by the total power generation, and the renewable energy is fully accommodated.

The optimized reactive power output of compensation device SVC and CB and OLTC optimal ratio are shown in Figure 3. The tap ratio of OLTC was changed four times, and the SVC and CB were changed frequently to provide sufficient reactive power. The system nodal voltage profile is shown in Figure 4, where all the nodal voltages in the range of [0.96, 1.04] ensure satisfied voltage security, and most of the voltages near 1 p.u. are with very small voltage deviations. These demonstrated that effective regulation of the reactive/voltage was achieved by optimally controlling the SVC and CB and OLTC, and thus a relatively good system voltage security was obtained.

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

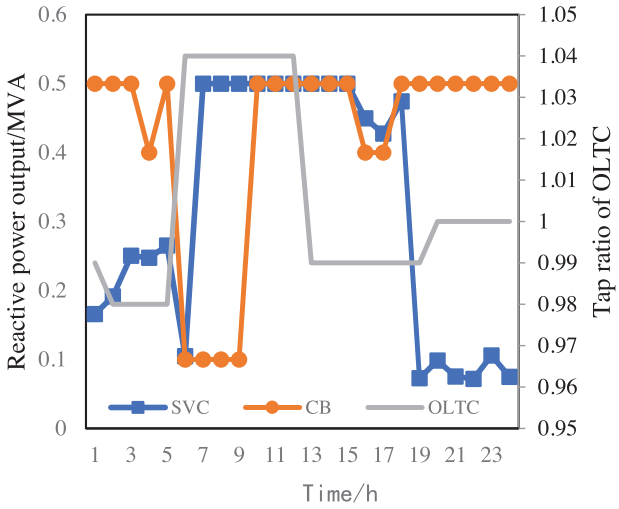


FIGURE 3 Reactive power output of SVC and CB

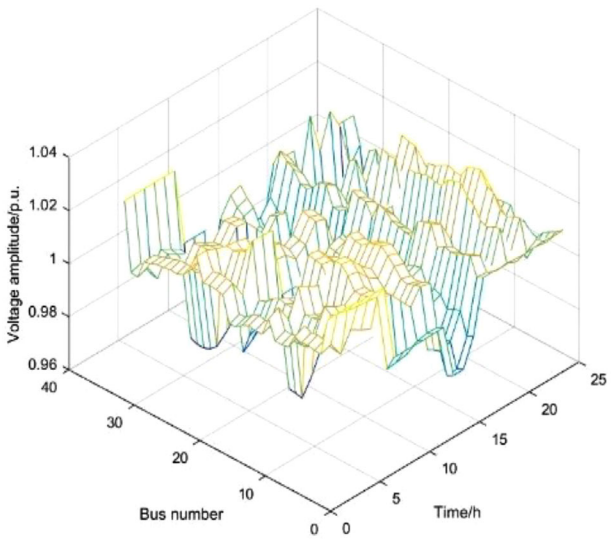


FIGURE 4 Nodal voltage profile of ADN

Figure 5 shows the SOP active and reactive power output obtained from the model. It is evident that the network loss could be reduced by SOP, and simultaneously the reactive power of the branch installed with SOP is properly controlled by SOP to ensure the nodal voltage. The observation that the active power and reactive power of the branch where the SOP is located is greatly affected by the change of the network topology indicated the coordination role of SOP for dynamic reconfiguration.

4.3 | Detailed analysis of SOP locations

To analyze the influence of SOP and SOP access location on the coordinated operation results of the distribution network, this paper compares the case without SOP installation and the case with SOP installed at different nodes, and the data is

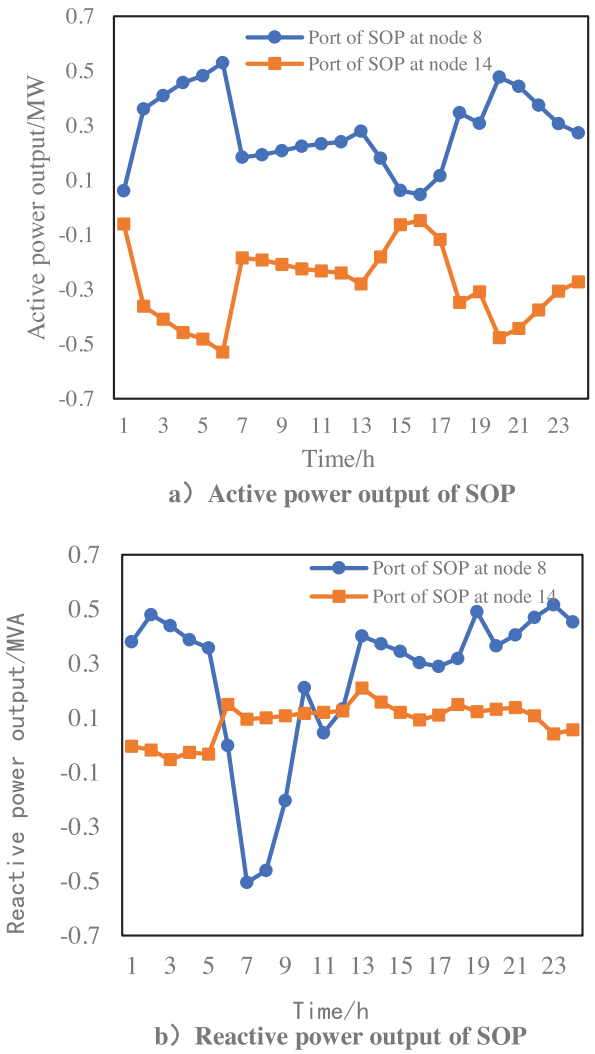


FIGURE 5 Active and reactive power output of SOP

shown in Table 4. Taking the operation optimization results of the case without SOP installation as a reference, when SOP is connected to the branch 8–14, the daily operating cost is reduced by 2546 yuan, and the average node voltage deviation is reduced by 3.25%; when SOP is connected to the branch 7–20, the daily operating cost is reduced by 2102 yuan, and the average value of node voltage deviation is reduced by 2.44%. This means that when comparing the case of SOP installed on the branch 8–14 with that of SOP installed on the branch 7–20, the former further reduced the daily operating cost of 444 yuan, and the corresponding average node voltage deviation was also smaller. It can be concluded that SOP can effectively reduce the operating cost of the distribution network and optimize the system nodal voltage.

4.4 | Cost comparisons of different simulation scenarios

In order to fully explore the effects of SVC, CB, reactive power/voltage control equipment, ESS, MGT, and SOP

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 Cost and voltage comparisons of cases with and without SOP.

SOP connection location	Total cost	Daily operation cost (Yuan)			Mean of nodal voltage deviation (p.u.)
		MGT generation cost	Power purchase cost	Penalty cost of voltage deviation	
No SOP	26795	13389	13314	92	0.0123
8–14	24249	13389	10811	48	0.0099
7–20	24693	14077	10544	72	0.0120

TABLE 5 Operating costs of distribution networks for different scenarios.

Scenario	Total cost	Daily operation cost (Yuan)		
		MGT power generation cost	Power purchase cost	Voltage deviation penalty cost
1	24249	13389	10811	48
2	31918	0	31502	415
3	30732	0	30591	141
4	26795	13389	13314	92
5	25754	13626	12128	186

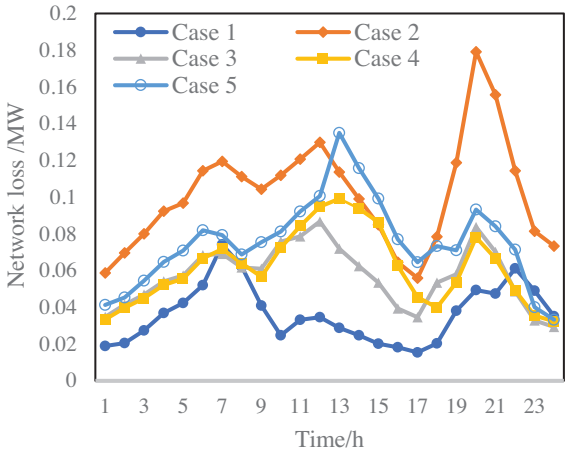


FIGURE 6 Network loss of different scenarios

equipment on the optimal operation of the distribution network, the following six different scenarios are settled based on the distributed renewable energy penetration rate of 60% to conduct the comparative analysis.

Scenario 1: Network dynamic reconfiguration was performed with contact switches, MGT, ESS, reactive power/voltage control devices, and SOP devices.

Scenario 2: Passive distribution network without any reconfiguration devices.

Scenario 3: Network dynamic reconfiguration was performed based on contact switches only.

Scenario 4: Network dynamic reconfiguration was performed based on contact switches and MGT.

Scenario 5: Network dynamic reconfiguration was performed based on contact switches, MGT and ESS.

Scenario 6: Network dynamic reconfiguration was performed based on contact switches, MGT, ESS, and reactive power/voltage control devices.

1. Operation costs comparisons of different scenarios

To compare the impact of dynamic reconfiguration strategy, MGT and ESS regulation strategy on the distribution network operation cost, Scenarios 1–5 were selected for comparison. The daily operation cost obtained is shown in Table 5, and the network loss is shown in Figure 6. Compared with Scenario 2, the system network loss of Scenario 3 is significantly lower in all time periods. The daily operating cost of Scenario 3 is reduced by 1,186 Yuan (3.71%) compared with that in Scenario 2, and the voltage deviation penalty cost is reduced by 274 Yuan. The results indicate that dynamic reconfiguration of the distribution network can effectively reduce the system network loss and thus, reduce the system's daily operation cost. Compared with Scenario 3, the daily operating cost of the distribution system is reduced by 3,937 Yuan in Scenario 4 by generating electricity with micro gas turbines. Compared with Scenario 4, the distribution system operating cost is reduced by 1,041 Yuan (3.89%) in Scenario 5 through ESS charging and discharging regulation. The results show that it is possible to reduce the distribution system operation cost with the use of ESS. In Scenario 1, DG output regulation, network dynamic reconfiguration, reactive/voltage control device regulation and SOP operation lead to coordinated optimization and obtain minimum daily operating cost and network loss. The results demonstrate that the proposed ADN day-ahead active and reactive power coordinated optimization second-order cone model could comprehensively explore the dynamic reconfiguration technology with multiple flexibly controllable resources in ADN to obtain the best day-ahead dispatch results, and thus these results verify the effectiveness of the proposed model in improving the operating economy of the distribution system.

2. Nodal voltage comparisons of different scenarios

As the penetration rate of renewable energy increases, the ADN voltage violation problem is growing in severity, which necessitates the introduction of reactive power/voltage control devices and SOPs to regulate the voltage level of distribution networks. In this section, the aforementioned scenarios 1, 5,

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

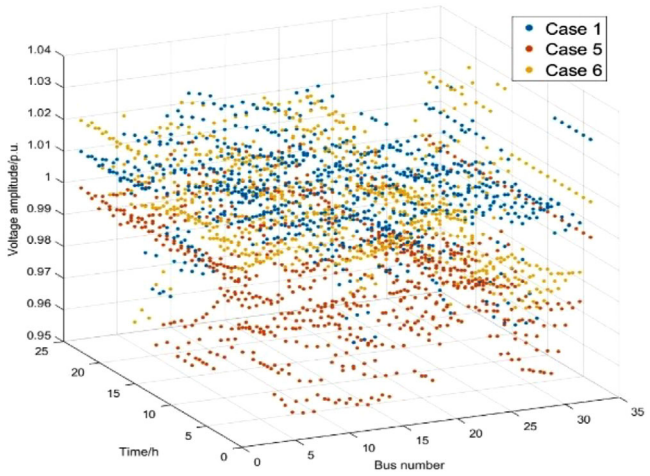


FIGURE 7 System temporal and spatial nodal voltages in scenarios 1, 5, and 6

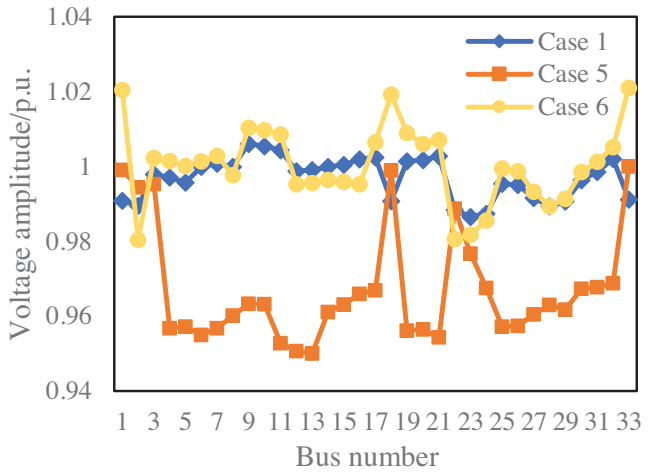


FIGURE 8 System nodal voltages during 14:00–15:00 in scenarios 1, 5, and 6

and 6 are selected for comparative analysis in order to analyze the effects of reactive power/voltage control devices and SOPs on distribution network nodal voltage, and the nodal voltage magnitudes of each scenario are shown in Figure 7. The mean value of the nodal voltage deviation is 0.0099 p.u. for Scenario 1, 0.0192 p.u. for Scenario 5, and 0.0123 p.u. for Scenario 6. Compared with Scenario 5, the nodal voltage deviations are restrained in Scenario 6 through the connection of reactive power/voltage control devices, which reduces the risk of voltage violations in the distribution system. Compared with Scenario 6, the nodal voltage deviation is further reduced in Scenario 1 and the system voltage level is improved through SOP coordinated regulation. These results show the effectiveness of the proposed method in Scenario 1 in improving the voltage profiles of the distribution system.

As shown in Figure 8, most of the nodes in Scenario 5 are very near the boundary 0.95 p.u. during 14:00–15:00 and thus the system voltage deviation is very large. Furthermore, as the voltage of node 13 has reached the lowest value of 0.95 p.u

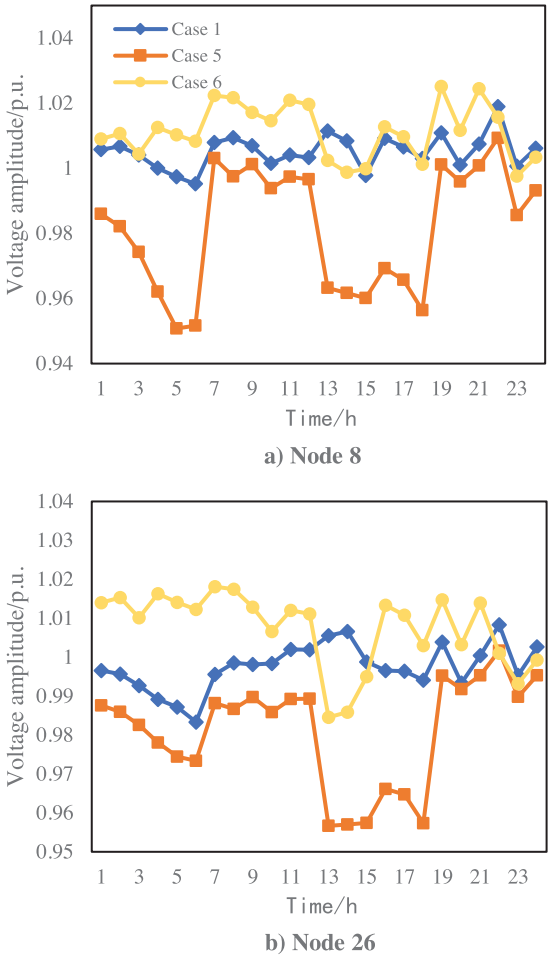


FIGURE 9 Voltage profiles of nodes 8 and 26 in scenarios 1, 5, and 6

meanwhile the voltage of node 1 is 1 p.u. as the highest one, there is a large voltage difference among the 33 nodes. Compared with Scenario 5, the node voltage deviation is significantly reduced and the system voltage level is improved for Scenario 6 where the reactive power compensation device is considered. Moreover, when the SOP is optimally coordinated with the network dynamic reconfiguration by switches, MGT, ESS, and the reactive power/voltage control devices in Scenario 1, the node voltage deviation is further reduced and the voltage difference among 33 nodes is very small. These results show that the reactive power/voltage control device and SOP device can effectively equalize the voltage level between nodes.

The 24-h voltage curves of node 8 (SOP connection node) and node 26 (SVC installation node) for scenarios 1, 5, and 6 are shown in Figure 9. Compared with Scenario 5, a significant reduction of voltage deviation and a decrease in voltage fluctuation are observed in Scenario 6, and the nodal voltage daily characteristic is improved. Compared with Scenario 6, the optimized voltage profile of Scenario 1 is in the range of [0.985, 1.010] which means the maximum voltage deviation is 0.015 p.u with regard to the per unit value. The voltage fluctuations during 24 h are also very small in Scenario 1. These results and analysis show that the coordinated optimization of SOP

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and reactive power/voltage control devices could well control the nodal voltage near rated voltage and effectively alleviate the system voltage fluctuations.

5 | CONCLUSIONS

In this paper, distributed generation units including MGT and ESS, the reactive power compensation devices SVC and CB, distribution network dynamic reconfiguration, and SOP devices are coordinated for the optimal operation of active distribution networks penetrated with renewables, and a day-ahead active/reactive power coordinated optimization model is established as a second-order cone problem to minimize the daily operation cost of ADN. The following conclusions could be drawn based on the case studies and discussions.

1. The proposed ADN day-ahead active and reactive power coordinated optimization model can realize dynamic reconfiguration of the distribution network and could be coordinated with multiple controllable resources. Compared with just one optimization measure, the proposed coordinated optimization model would substantially improve the voltage quality and reduce the system operation cost, thus providing a safe and economic dispatching scheme for ADN operation.
2. Dynamic reconfiguration of the distribution network by contact switches would reduce the daily operating costs of ADN by 3.71% compared with the passive distribution network without any reconfiguration devices; if the ESS is further coordinated with contact switches, the daily operating costs of ADN would additionally reduce 3.98%, meanwhile, the voltage deviation and system voltage fluctuation can be alleviated by the coordination of reactive power/voltage control devices.

In the paper, the proposed active and reactive power coordinated optimization of ADN does not consider the uncertainties of renewable power generation, and in the future a stochastic multiple time-scale coordinated optimization model would be established for the economic and reliable operation of ADN with highly penetrated uncertain renewables.

AUTHOR CONTRIBUTIONS

Chen Li: Writing—review & editing. Yao Dai: Writing—review & editing. Peng Wang: Writing—review & editing. Shiwei Xia: Writing—review & editing

CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

No.

ORCID

Shiwei Xia https://orcid.org/0009-0007-2734-1269

REFERENCES

1. Hu, J., et al.: Optimal operation of energy hubs with large-scale distributed energy resources for distribution network congestion management. IEEE Trans. Sustainable Energy. 12(3), 1755–1765 (2021)
2. McDonald, J.: Adaptive intelligent power systems: active distribution networks. Energy Policy 36(12), 4346–4351 (2008)
3. Xia, S., et al.: A fully distributed hierarchical control framework for coordinated operation of DERs in active distribution power networks. IEEE Trans. on Power Syst. 34(6), 5184–5197 (2019)
4. Ge, X., et al.: Optimal day-ahead scheduling for active distribution network based on improved information gap decision theory. IET Renewable Power Gener. 15, 952–963 (2021)
5. Xu, B., et al.: Reactive power optimization of a distribution network with high-penetration of wind and solar renewable energy and electric vehicles. Prot. Control Mod. Syst. 7(4), 801–813 (2022)
6. Santos, S.F., et al.: Dynamic distribution system reconfiguration considering distributed renewable energy sources and energy storage systems. IEEE Syst. Journal 16(3), 3723–3733 (2022)
7. Zhou, A., et al.: Three-phase unbalanced distribution network dynamic reconfiguration: A distributionally robust approach. IEEE Trans. Smart Grid 13(3), 2063–2074 (2022)
8. Shariatzadeh, F., et al.: Real-time implementation of intelligent reconfiguration algorithm for microgrid. IEEE Trans. Sustainable Energy 5(2), 598–607 (2014)
9. Wang, B., et al.: Distribution network reconfiguration based on noisynet deep q-learning network. IEEE Access 9, 90358–90365 (2021)
10. Qi, Q., Wu, J.: Increasing distributed generation penetration using network reconfiguration and soft open points. Energy Proc. 105, 2169–2174 (2017)
11. Azizi, A., Vahidi, B., Nematollahi, A.F.: Reconfiguration of active distribution networks equipped with soft open points considering protection constraints. J. Mod. Power Syst. Clean Energy 11(1), 212–222 (2023)
12. Hieba, A.A.: A novel parallel quntum genetic optmization techniuqe with soft open point for distributed generators electrical network reconfiguration. In: 2022 International Telecommunications Conference (ITC-Egypt), pp. 1–4. IEEE, Piscataway, NJ (2022)
13. Liu, Y., Li, J., Wu, L.: Coordinated optimal network reconfiguration and voltage regulator/DER control for unbalanced distribution systems. IEEE Trans. Smart Grid 10(3), 2912–2922 (2019)
14. Home-Ortiz, J.M., et al.: Optimal operation of active distribution systems with voltage control and closed-loop topology. In: 2021 IEEE Madrid PowerTech, pp. 1–6. IEEE, Piscataway, NJ (2021)
15. Fetouh, T., Elsayed, A.M.: Optimal control and operation of fully automated distribution networks using improved tunicate swarm intelligent algorithm. IEEE Access 8, 129689–129708 (2020)
16. Peng, Q., Tang, Y., Low, S.H.: Low: Feeder reconfiguration in distribution networks based on convex relaxation of OPF. IEEE Trans. Power Syst. 30(4), 1793–1804 (2015)
17. Nikkhah, S., et al.: Risk averse energy management strategy in the presence of distributed energy resources considering distribution network reconfiguration: an information gap decision theory approach. IET Renewable Power Gener. 14, 305–312 (2020)
18. Ahmadi, H., Martí, J.R.: Mathematical representation of radiality constraint in distribution system reconfiguration problem. Int. J. Electr. Power Energy Syst. 64, 293–299 (2015)
19. Wang, J., et al.: Reducing the active power curtailment of distributed generation based on soft open points in unbalanced distribution networks. IET Renewable Power Gener. 14, 3252–3258 (2020)
20. Li, P., et al.: Multi-stage expansion planning of energy storage integrated soft open points considering tie-line reconstruction. Prot. Control Mod. Power Syst. 7(4), 683–697 (2022)
21. Jiang, X., et al.: An overview of soft open points in electricity distribution networks. IEEE Trans. Smart Grid 13(3), 1899–1910 (2022)
22. Liu, H., et al.: Distributionally robust co-optimization of the demand-side resources and soft open points allocation for the high penetration of renewable energy. IET Renewable Power Gener. 16, 713–725 (2022)

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

23. Baran, M.E., Wu, F.F.: Network reconfiguration in distribution systems for loss reduction and load balancing. IEEE Trans. Power Delivery 4(2), 1401–1407 (1989)
24. Qiu, Y., et al.: Adaptive uncertainty sets-based two-stage robust optimisation for economic dispatch of microgrid with demand response. IET Renewable Power Gener. 14, 3608–3615 (2020)
25. Lu, X., et al.: A model for balance responsible distribution systems with energy storage to achieve coordinated load shifting and uncertainty mitigation. Energy 249, 123749 (2022)
26. Agalgaonkar, Y.P., Pal, B.C., Jabr, R.A.: Distribution voltage control considering the impact of PV generation on tap changers and autonomous regulators. IEEE Trans. Power Syst. 29(1), 182–192 (2014)
27. Jabr, R.A., Singh, R., Pal, B.C.: Minimum loss network reconfiguration using mixed-integer convex programming. IEEE Trans. Power Syst. 27(2), 1106–1115 (2012)
28. Xia, S., et al.: Multitime scale coordinated scheduling for the combined system of wind power, photovoltaic, thermal generator, hydro pumped storage, and batteries. IEEE Trans. Industry Appl. 56(3), 2227–2237 (2020)
29. Sheibani, M.R., et al.: Energy storage system expansion planning in power systems: a review. IET Renewable Power Gener. 12, 1203–1221 (2018)

How to cite this article: Li, C., Dai, Y., Wang, P., Xia, S.: Active and reactive power coordinated optimization of active distribution networks considering dynamic reconfiguration and SOP. IET Renew. Power Gener. 19, 1–12 (2025). https://doi.org/10.1049/rpg2.12814

APPENDIX A

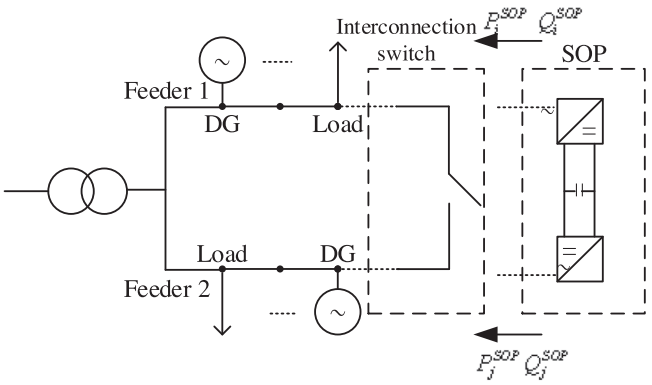


FIGURE A1 Installation of SOP

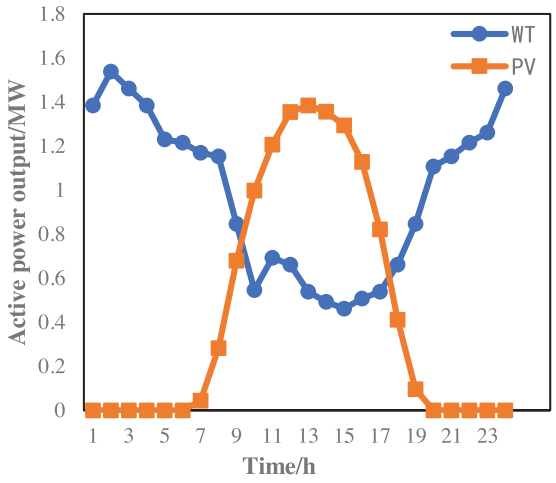


FIGURE A2 Predicted renewable energy output

17521424, 2025, 1, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/rpg2.12814, Wiley Online Library on [09/11/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License